

Securitize Solana Redemption Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

Farouk

Naman

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Securitize on Securitize Solana Redemption. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated August 29, 2025, was delivered to Securitize, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 5 days, the Cyfrin team conducted an audit on the Securitize Solana Redemption code provided by Securitize. The findings consist of 1 High, 2 Medium & 1 Low severity issues with the remainder being informational.

Summary

Project Name	Securitize Solana Redemption
Repository	bc-solana-redemption-sc
Commit	36051ebd2739...
Audit Timeline	Aug 19th - Aug 25th, 2025
Methods	Manual Review

Issues Found

Severity	Found	Resolved	Acknowledged
Critical Risk	0	0	0
High Risk	1	1	0
Medium Risk	2	2	0
Low Risk	1	1	0
Informational	3	2	1
Gas Optimizations	0	0	0
Total	7	6	1

6 Audit Scope

Cyfrin conducted an audit of the Securitize Solana Redemption program based on the code present in the repository commit hash 36051eb. The following files construct the scope of the audit:

```
programs/securitize-off-ramp/src/constants.rs
programs/securitize-off-ramp/src/errors.rs
programs/securitize-off-ramp/src/events.rs
programs/securitize-off-ramp/src/instructions/calculate_liquidity_token_amount.rs
programs/securitize-off-ramp/src/instructions/change_admin.rs
programs/securitize-off-ramp/src/instructions/change_liquidity_withdraw_authority.rs
programs/securitize-off-ramp/src/instructions/initialize.rs
programs/securitize-off-ramp/src/instructions/mod.rs
programs/securitize-off-ramp/src/instructions/pause.rs
programs/securitize-off-ramp/src/instructions/redeem.rs
programs/securitize-off-ramp/src/instructions/unpause.rs
programs/securitize-off-ramp/src/instructions/update_asset_policy.rs
programs/securitize-off-ramp/src/instructions/update_countries_restriction.rs
programs/securitize-off-ramp/src/instructions/update_fee_manager.rs
programs/securitize-off-ramp/src/instructions/update_nav_provider.rs
programs/securitize-off-ramp/src/instructions/withdraw_liquidity.rs
programs/securitize-off-ramp/src/lib.rs
programs/securitize-off-ramp/src/states/mod.rs
programs/securitize-off-ramp/src/states/off_ramp_counter.rs
programs/securitize-off-ramp/src/states/off_ramp_state.rs
programs/securitize-off-ramp/src/utils/fee_manager/fee_manager_core.rs
programs/securitize-off-ramp/src/utils/fee_manager/mod.rs
programs/securitize-off-ramp/src/utils/fee_manager/mpbs_fee_manager.rs
programs/securitize-off-ramp/src/utils/get_authority_signer.rs
programs/securitize-off-ramp/src/utils/get_rate.rs
programs/securitize-off-ramp/src/utils/get_transfer_hook_program_id.rs
programs/securitize-off-ramp/src/utils/mod.rs
programs/securitize-off-ramp/src/utils/token_calculator.rs
programs/securitize-off-ramp/src/utils/transfer_maybe_hook.rs
```

7 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

8 Protocol Summary

Securitize Off-Ramp is a Solana program that converts an asset token into a liquidity token using a NAV rate, with identity checks, country restrictions and fees. It is composed of:

- **State**
  - `OffRampCounter`: global counter for unique off-ramp ids.
  - `OffRampState`: per-instance configuration, including the asset mint, the asset policy (Recipient or Burn), the liquidity mint, the NAV provider program id, the fee manager and the restricted countries.
- **Integrations**
  - **Identity stack**: `IdentityRegistryAccount`, `IdentityAccount` and `WalletIdentity` associate wallets with identities and their country codes.
  - **NAV provider program**: CPI interface that returns a rate for the asset mint.
  - **SPL Token and Token-2022**: asset and liquidity mints may be legacy or Token-2022, with optional extensions and transfer hooks.
- **Core instructions**
  - **Initialize**: creates an `OffRampState`, increments the counter and creates the liquidity vault.
  - **Redeem**: fetches the NAV rate, computes the output and fee, moves liquidity tokens to the user and fee collector, and burns or deposits the asset according to the asset policy.
  - **Withdraw liquidity**: transfers liquidity tokens out of the vault.
  - **Configuration updates**: update the off-ramp configuration.
  - **Quote**: returns the computed liquidity amount for a given asset amount, optionally before fees.

9 Attestation

Cyfrin attests that it performed a security review of the Securitize Solana Redemption code provided by Securitize.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2025-08-29-cyfrin-securitize-solana-redemption-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Securitize has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=f5819430aa2dfd56ca92d3b94aace2977e5fdd01f40a384b0caca505f880aed0
F=2025-08-29-cyfrin-securitize-solana-redemption-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.